

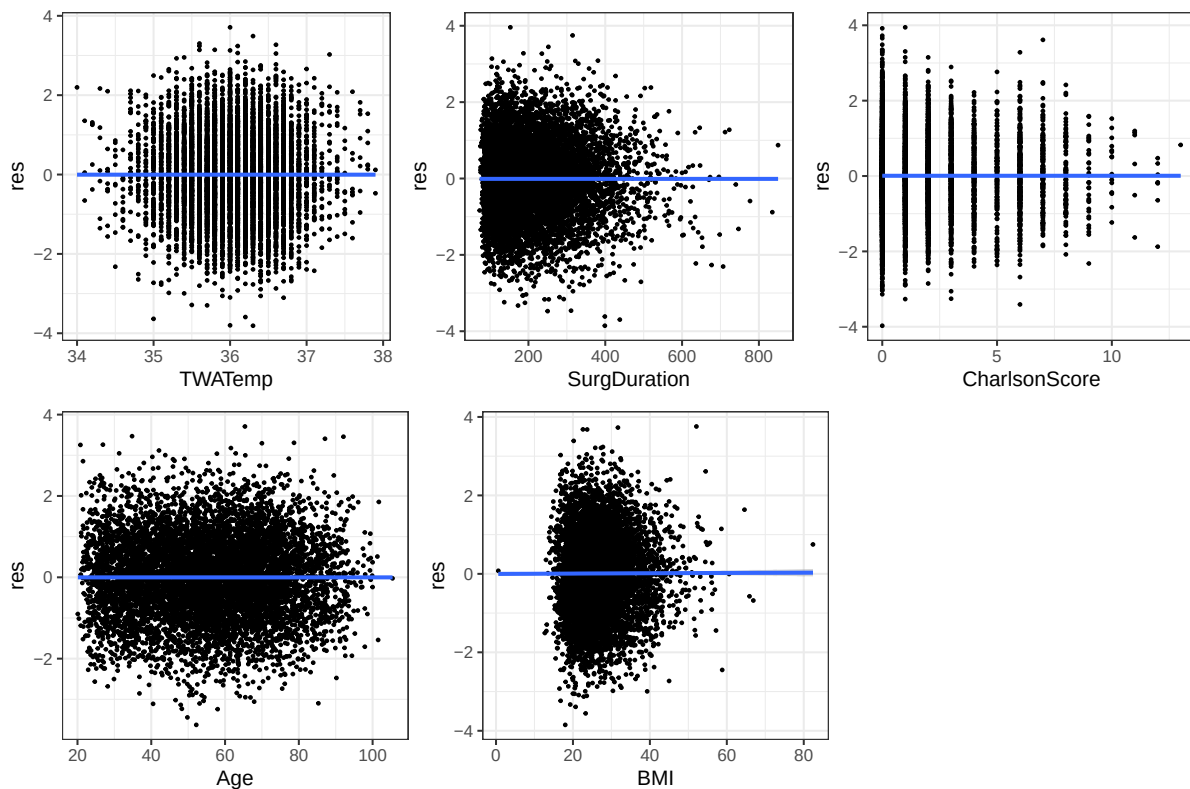
# Model 1

## Model 1

```
m1 <- glm(SeriousInfection ~ TWATemp + SurgDuration + Open +  
  Age + FEMALE + BMI + CharlsonScore + CHF + VALVE +  
  RENLFAIL + LIVER + METS + TUMOR + COAG +  
  OBESE + LYLES + BLDLOSS + ANEMDEF + DRUG +  
  SteroidHx + ImmunosuppressantHx,  
  data = ct, family = binomial)
```

## Diagnostics

Quantile residual plots:



Cook's distance:

```
summary(cooks.distance(m1))
```

| Min.      | 1st Qu.   | Median    | Mean      | 3rd Qu.   | Max.      |
|-----------|-----------|-----------|-----------|-----------|-----------|
| 7.554e-07 | 2.686e-06 | 6.252e-06 | 1.296e-04 | 1.915e-05 | 7.586e-03 |

Variance inflation factors:

```
car::vif(m1)
```

|                     | TWATemp  | SurgDuration | Open          | Age       |
|---------------------|----------|--------------|---------------|-----------|
|                     | 1.073678 | 1.073921     | 1.042975      | 1.391585  |
|                     | FEMALE   | BMI          | CharlsonScore | CHF       |
|                     | 1.034951 | 1.670663     | 7.177541      | 1.454919  |
|                     | VALVE    | RENLFAIL     | LIVER         | METS      |
|                     | 1.285152 | 1.783144     | 1.152668      | 4.091548  |
|                     | TUMOR    | COAG         | OBESE         | LYTES     |
|                     | 2.050034 | 1.173009     | 1.614425      | 1.165845  |
|                     | BLDLOSS  | ANEMDEF      | DRUG          | SteroidHx |
|                     | 1.024956 | 1.110433     | 1.043339      | 1.091266  |
| ImmunosuppressantHx | 1.041512 |              |               |           |

## Model summary

```
summary(m1)
```

Call:

```
glm(formula = SeriousInfection ~ TWATemp + SurgDuration + Open +  
  Age + FEMALE + BMI + CharlsonScore + CHF + VALVE + RENLFAIL +  
  LIVER + METS + TUMOR + COAG + OBESSE + LYTES + BLDLOSS + ANEMDEF +  
  DRUG + SteroidHx + ImmunosuppressantHx, family = binomial,  
  data = ct)
```

Coefficients:

|             | Estimate   | Std. Error | z value | Pr(> z )    |
|-------------|------------|------------|---------|-------------|
| (Intercept) | -7.4822782 | 2.8566915  | -2.619  | 0.008813 ** |

|                     |            |           |        |          |     |
|---------------------|------------|-----------|--------|----------|-----|
| TWATemp             | 0.1175861  | 0.0792734 | 1.483  | 0.137995 |     |
| SurgDuration        | 0.0030224  | 0.0003642 | 8.298  | < 2e-16  | *** |
| Open                | 0.1915301  | 0.1184500 | 1.617  | 0.105885 |     |
| Age                 | -0.0064543 | 0.0028712 | -2.248 | 0.024581 | *   |
| FEMALE              | -0.0992990 | 0.0834218 | -1.190 | 0.233919 |     |
| BMI                 | 0.0020268  | 0.0085134 | 0.238  | 0.811829 |     |
| CharlsonScore       | 0.0670343  | 0.0500116 | 1.340  | 0.180123 |     |
| CHF                 | 0.2232918  | 0.2224313 | 1.004  | 0.315442 |     |
| VALVE               | -0.1783084 | 0.1867670 | -0.955 | 0.339724 |     |
| RENLFAIL            | 0.2209295  | 0.2162451 | 1.022  | 0.306941 |     |
| LIVER               | -0.4471262 | 0.2741792 | -1.631 | 0.102936 |     |
| METS                | -0.6086990 | 0.3435674 | -1.772 | 0.076444 | .   |
| TUMOR               | -0.2157553 | 0.1531927 | -1.408 | 0.159015 |     |
| COAG                | 0.4779159  | 0.1316557 | 3.630  | 0.000283 | *** |
| OBESE               | -0.2273914 | 0.1407603 | -1.615 | 0.106213 |     |
| LYTES               | 0.6058007  | 0.0890399 | 6.804  | 1.02e-11 | *** |
| BLDLOSS             | 0.2001911  | 0.2349857 | 0.852  | 0.394254 |     |
| ANEMDEF             | 0.0373394  | 0.0966725 | 0.386  | 0.699314 |     |
| DRUG                | 0.5353883  | 0.2238786 | 2.391  | 0.016783 | *   |
| SteroidHx           | 0.0639630  | 0.0888817 | 0.720  | 0.471745 |     |
| ImmunosuppressantHx | -0.0855607 | 0.2312001 | -0.370 | 0.711329 |     |

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Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 4608.2 on 7907 degrees of freedom  
 Residual deviance: 4393.1 on 7886 degrees of freedom  
 AIC: 4437.1

Number of Fisher Scoring iterations: 5